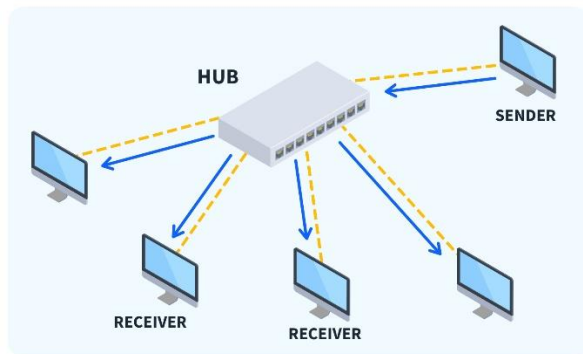


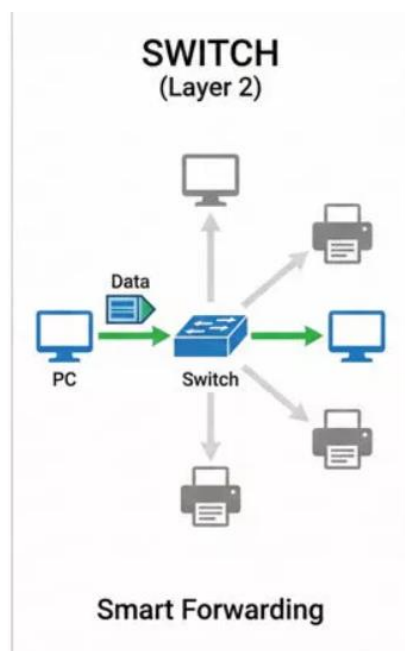
Before you directly jump into these question..... here is a smolll definitions. That might me helpful like for better understanding... iq

HUB: A Hub is a basic networking device used to connect multiple computers in a network.

(Like a extension box we can connect multiple devices....)



SWITCH: A Switch is a networking device that connects devices in a LAN (means local area network.... Which covers a small area like home) and forwards data only to the destination device using MAC addresses.



HUB vs SWITCH

FEATURE	HUB	SWITCH
SECURITY	low	Better than hub
SPEED	slower	faster
DATA TRANSMISSION	Broadcasts (sends) to all devices	Sends to specific device.
OSI LAYER	Physical layer	Data link layer
DUPLEX MODE	Half-duplex	Full duplex

These are just key words... u've to make a sentence like hub's security is low. Like that!!!

STAR TOPOLOGY

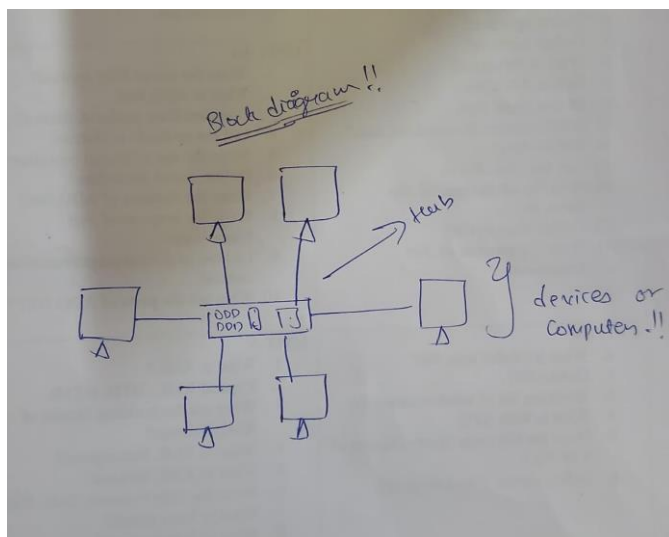
Again before jumping into the topic just a smoll info..

TOPOLOGY:

- it's a physical layout of LAN network.
- It includes cables etc.

STAR TOPOLOGY

Block diagram.(this is enough but don't forget to mention hub, and computers like... comp1, comp2 It's ur wish.. u can take any one of the devices as comp1)



- It is a network In which all the devices are connected to a central device called HUB.
- When a device wants to send a data to other devices, it sends the data to the hub and the hub passes the data to other devices.... (like a mediator)
- The deevices r not directly connected to each other.
- Doesn't allow network traffic b/w the devices

ADVANTAGES:

- Easy to install
- Easy to configure
- Less expensive
- Easy to detect faults

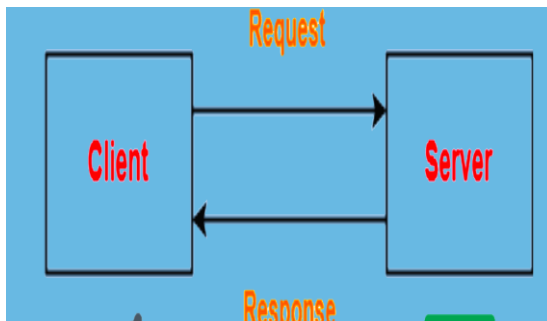
DISADVANTAGES:

- If the hub fails, the entire network will be a failure
- Need more cable length. (optional... coz disavd la one is enough!!)

PROTOCOLS

1. **HTTP**: it stands for hyper text transfer protocol

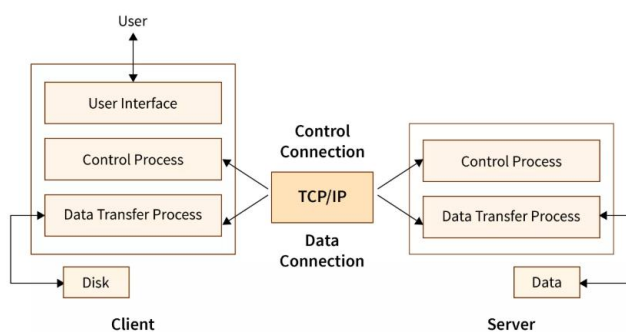
Block diagram.



HTTP is an application layer protocol used to transfer web pages and data between a client (browser) and a server.

Example: open a website in chrome... (like www.youtube.com etc)

2. **FTP**: it stands for file transfer protocol.(block diagram is bit confusing)



FTP is an application layer protocol used to transfer files between computers over a network.

Example: Uploading files to a website server

3. **VoIP**: it stands for voice over internet protocol.

Iska block diagram idk.... ;-;

VoIP is a protocol that allows voice communication over the internet instead of traditional telephone lines.

Example : whatsapp call, zoom meeting etc.